

Ishaan Gupta

ML ENGINEER | DEEP LEARNING RESEARCHER

San Diego / Bay Area, CA

✉ i3gupta@ucsd.edu | 🏠 ishaanSD.github.io/home | 📧 IshaanSD | 🌐 IgAI

ML PhD researcher in deep learning applications, building robust AI pipelines for noisy, limited-label real-world data, with experience spanning biotechnology, surgical analytics, recommendation systems, and data engineering.

Education

University of California, San Diego

La Jolla, CA

Ph.D. Computer Science — Deep Learning for Genomics | M.S. Computer Science, earned en route

Sep 2023 – Mar 2028 (exp.)

• M.S. awarded Jun 2025 | GPA 3.92 | Charles Lee Powell Research Fellowship | Advisors: Tiffany Amariuta, Pavel A. Pevzner

University of California, San Diego

La Jolla, CA

B.S. Computer Science — Bioinformatics Specialization

Aug 2019 – Jun 2022

• GPA 3.92 | Magna Cum Laude | IEEE Eta Kappa Nu | Minor in Biology

Experience

Halıcıoğlu Data Science Institute, UCSD

San Diego, CA

Graduate Student Researcher — Deep Learning (Dr. Tiffany Amariuta)

Sep 2024 – Present

- Built *DiploFormer*, a Nextflow/**PyTorch** framework to adapt foundation models to personal genomes for disease mechanisms discovery.
- Fine-tuned **186M-parameter** FlashAttention **transformer** + CNN-based model across people and sequence contexts on multi-modal data.
- Implemented delayed pairwise-loss computation via output accumulation, ran **WandB** / MLflow sweeps of adaptation strategies and ablations.
- Optimized resource-aware CPU/GPU workflows with parallel jobs, model checkpoint recovery across preprocessing, training, and inference.
- Benchmarked against Elastic Net/**XGBoost** baselines, monitored data drift, verified discovery of causal features through external validation.

Intuitive Surgical, Inc.

Sunnyvale, CA

AI Data Scientist Intern

Jun 2026 – Sep 2026

- Engineered Python/SQL components across the full analytics pipeline, transforming action-recognition outputs from 100K+ surgical cases into validated event tables, benchmark metrics, and dashboard-ready **user-facing insights** for analyzing hospital operational efficiency.
- Improved **temporal analytics** robustness by evaluating **outlier-detection** strategies and implementing a custom GMM/IQR detector.
- Built an LLM-assisted data-ingestion and validation workflow to standardize inconsistent hospital observation spreadsheets for user research
- Enabled detector robustness and longitudinal trend analyses that identified service-associated improvements in business outcomes.

Abterra Biosciences

San Diego, CA

Machine Learning Engineer Intern

Jul 2024 – Sep 2024

- Benchmarked autoregressive transformer models for de novo peptide sequencing, improving held-out performance **4x** on in-house data.
- Built reproducible PyTorch Lightning + W&B training workflows for hyperparameter sweeps, model comparison, and failure-mode analysis.
- Achieved **95% accuracy** (baseline: 60%) in resolving ambiguous antibodies through peptide-spectrum feature engineering and FNN classifiers.
- Deployed trained deep learning models in the company's Java codebase using DL4J, enabling parallelized performance evaluation.

UCSD CSE Department

San Diego, CA

Graduate Student Researcher — Algorithms (Dr. Pavel A. Pevzner)

Jan 2022 – Jul 2024

- Co-Designed GenomeDecoder (**C++**) for finding 10K-1M long inexact repeats in 100M long sequences by iteratively simplifying de Bruijn graphs
- Outperformed SOTA on segmental duplication detection in immune loci; Browser-friendly genome-scale visualization of the duplications *Bioinformatics* (2025).
- Improved UniAligner (suffix-array-based alignment) runtime **4-10x** and integrated with GenomeDecoder to discover gene deletions across primate genomes. *Nature* (2025)
- Developed signal-processing methods using cumulative sums, FFT, and clustering for delineating bidirectionality in noisy Nanopore signals.

Illumina, Inc.

San Diego, CA

Software Engineer I — Systems Integration

Jun 2022 – Dec 2022

- Developed production-grade ETL infrastructure code for high-throughput lab automation workflows processing **10,000s of samples/day**.
- Built REST APIs and event-driven microservices using Java Spring/JDBC, SQL databases, RabbitMQ, AWS S3/SQS, and TypeScript.
- Deployed **Dockerized** services on **Kubernetes** with SQL stored procedures, JUnit tests, and CI/CD release workflows.

Selected Projects

ChemBot - Agentic RAG Assistant for Trusted OTC Drug Information

LangGraph · MCP · RAG · ChromaDB · Google Cloud

(github.com/IshaanSD/ChemBot)

- Agentic system for over-the-counter drug information retrieval (eg: side-effects) grounded in trusted biomedical evidence (FAERS and PubMed).
- **LangGraph** routing across **MCP servers** integrated with RAG (ChromaDB), embeddings (SentenceTransformer), and LLMs (OpenAI/Gemini)
- Deployed on Google Cloud with automated evaluations for tool-calling, agent skills, citation faithfulness, and cross-LLM response quality.

Generative Models for Medical Image Augmentation

Diffusion Model · β -VAE · Context U-Net · GAN

(github.com/IshaanSD/DLCXR)

- Built a **diffusion model** training pipeline using β -VAE latent encoding and a Context U-Net denoiser, and demonstrated on chest X-ray images.
- Implemented disentangled β -VAE representations for CXR generation and a U-Net GAN baseline for mammography augmentation.

MitoEdit — Lab-in-the-Loop Agent for Mitochondrial Research at St. Jude

MCP · LLM Agents · FastAPI · Docker · Python (github.com/xz-stjude/MitoEdit)

- Built an **MCP server** and FastAPI layer that lets bench scientists run TalenWF through typed, agent-callable tools. *(manuscript in preparation)*
- Dockerized a CI/CD-enabled Python workflow + UI for identifying mtDNA cancer variants editable by TALE and predicting bystander edits.

Graphazon - Natural Language Query-based Product Recommendation

LangChain · Knowledge Graphs · LightGBM / LambdaMART (github.com/IshaanSD/Graphazon)

- Built a product-recommendation pipeline on the Amazon ESCI dataset, using an LLM to parse queries into typed intents.
- Combined knowledge-graph, Node2Vec, and text embeddings with **LambdaMART** learn-to-rank optimized for Mean Reciprocal Rank.

Technical Skills

Languages	Python, Java, C++, SQL, Bash, R
ML / Deep Learning	PyTorch, Lightning, Transformers, CNN, LSTM, PEFT / LoRA, FlashAttention, scikit-learn, XGBoost, LightGBM, TensorFlow
MLE / MLOps	W&B, MLflow, Hugging Face, Docker, FastAPI, REST APIs, CI/CD, pytest/JUnit, model evaluation, drift analysis
Data & Infrastructure	Spark, Kafka, Airflow, Pandas, Polars, SQL optimization, MongoDB, AWS, GCP, Slurm, Nextflow, Kubernetes
GenAI / Retrieval	LangGraph, LangChain, MCP, RAG, tool-calling, ChromaDB, SentenceTransformers, agent evaluation,

Publications & Conferences

- *DiploFormer: Deep Learning framework for capturing Non-Additive Regulatory Effects on Gene Expression and Chromatin Accessibility*. Presented at Keystone AI in Molecular Biology, 2025; **ASHG**, 2026. (github.com/IshaanSD/DiploFormer_1000G_atac)
- *GenomeDecoder: inferring segmental duplications in repetitive genomic regions*. **Bioinformatics**, 2025.
- *Complete sequencing of ape genomes*. **Nature**, 2025. (ishaansd.github.io/iPlots)

Certifications

- **AI Agents:** Google AI Agents Capstone | **Data Engineering:** IBM SQL, ETL/Airflow/Kafka, Spark/Hadoop | **Backend:** Advanced Java Spring Integration Testing

Teaching Experience

UC San Diego	San Diego, CA
Teaching Assistant / Tutor	Sep 2020 – Jun 2022
• Courses: <i>CSE 181, CSE 182, BICD 140, BILD 4</i>; taught algorithms, machine learning concepts, statistics for research, and biological data analysis. • Supported instruction on graph algorithms, dynamic programming, HMMs, clustering, sequence analysis, databases, and hypothesis testing.	